

Memo

(1) Assessment Scheme: Assignments 20%, Mid-term exam 30%, Final exam 50%.

(2) The number of assignments is 6 or 7 in total.



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

General Chemistry CHM1001

Lecture 2

Periodic Properties of the Elements (Chapter 7)

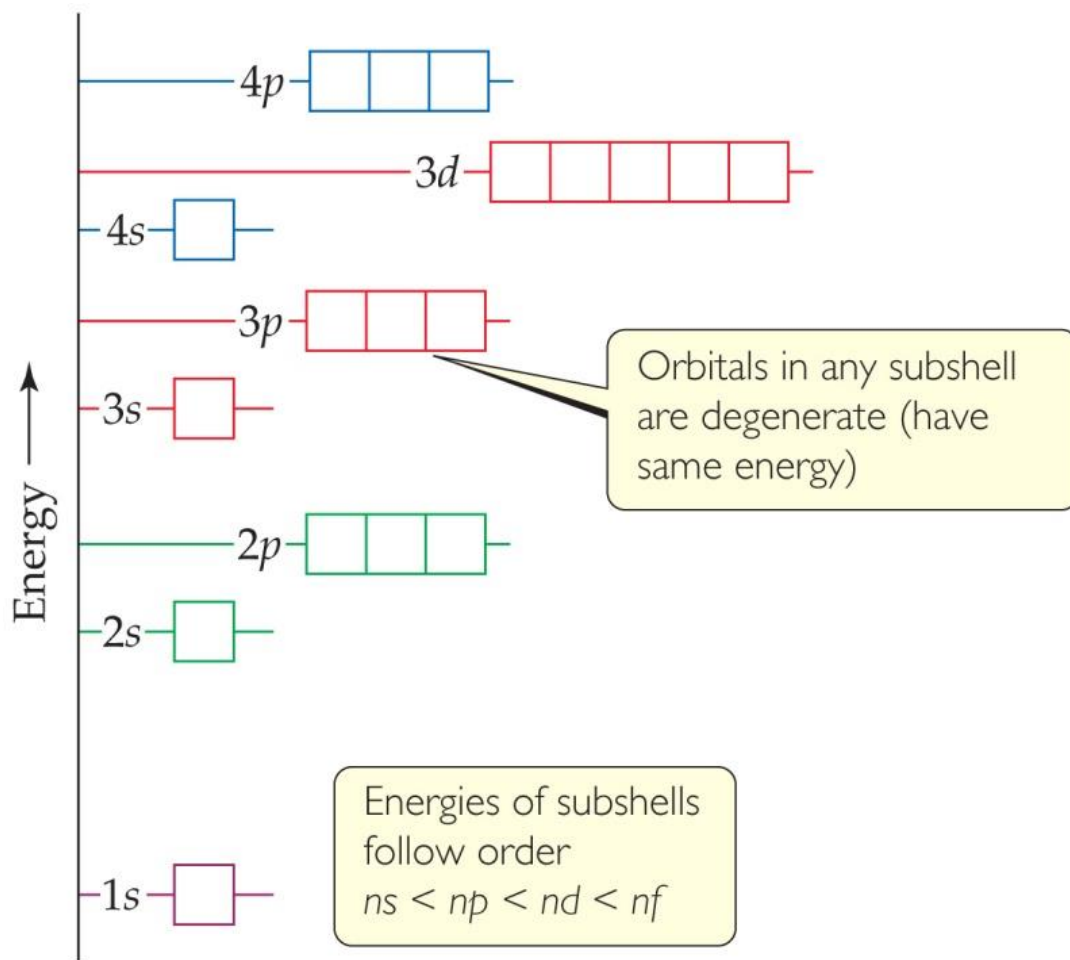
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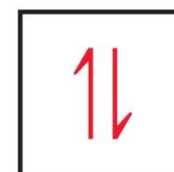
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Sep 5th, 2024

Recap



Li



1s



2s

Periodic
Properties
of the
Elements

Recap

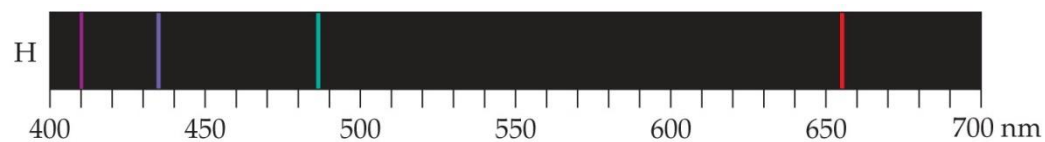
Atomic Emissions



Neon (Ne)



Hydrogen (H)



Recap

Table 6.3 Electron Configurations of Several Lighter Elements

Element	Total Electrons	Orbital Diagram					Electron Configuration
		1s	2s	2p			3s
Li	3	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow}$	$\boxed{}\boxed{}\boxed{}$	$\boxed{}$		$1s^2 2s^1$
Be	4	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow\downarrow}$	$\boxed{}\boxed{}\boxed{}$	$\boxed{}$		$1s^2 2s^2$
B	5	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow}\boxed{}\boxed{}$	$\boxed{}$		$1s^2 2s^2 2p^1$
C	6	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow}\boxed{\uparrow}\boxed{}$	$\boxed{}$		$1s^2 2s^2 2p^2$
N	7	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow}\boxed{\uparrow}\boxed{\uparrow}$	$\boxed{}$		$1s^2 2s^2 2p^3$
Ne	10	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}$	$\boxed{}$		$1s^2 2s^2 2p^6$
Na	11	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}$	$\boxed{\uparrow}$		$1s^2 2s^2 2p^6 3s^1$

Hund's rule

parallel $\rightarrow$ less energy $\rightarrow$ more stable

half

Development of the Periodic Table

H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	Lr	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	[113]	Fl	[115]	Lv	[117]	[118]

La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No

A horizontal timeline showing the history of the field of study. The timeline is divided into seven colored boxes, each representing a period. Below each box is the number of elements in that period.

Ancient Times	Middle Ages–1700	1735–1843	1843–1886	1894–1918	1923–1961	1965–
(9 elements)	(6 elements)	(42 elements)	(18 elements)	(11 elements)	(17 elements)	(15 elements)

Dmitri
Mendeleev and
Lothar Meyer
independently
came to the
same conclusion
about how
elements should
be grouped.

Mendeleev and the Periodic Table

Table 7.1 Comparison of the Properties of Eka-Silicon Predicted by Mendeleev with the Observed Properties of Germanium

Property	Mendeleev's Predictions for Eka-Silicon (made in 1871)	Observed Properties of Germanium (discovered in 1886)
Atomic weight	72	72.59
Density (g/cm^3)	5.5	5.35
Specific heat ($\text{J}/\text{g}\cdot\text{K}$)	0.305	0.309
Melting point ($^{\circ}\text{C}$)	High	947
Color	Dark gray	Grayish white
Formula of oxide	XO_2	GeO_2
Density of oxide (g/cm^3)	4.7	4.70
Formula of chloride	XCl_4	GeCl_4
Boiling point of chloride ($^{\circ}\text{C}$)	A little under 100	84

Chemists mostly credit Mendeleev because he also used chemical properties to organize the table and predicted some missing elements and their expected properties, including germanium.

Development of the Periodic Table

H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr

Periodic
Properties
of the
Elements

Atomic Number

- Mendeleev's table was based on atomic masses. It was the most fundamental property of elements known at the time.

1 1.0080
1 H 1 p⁺
Hydrogen
Nonmetal

2 4 9.012...
3 Li 7.0 4 Be
Lithium Alkali Metal Beryllium Alkaline Earth Metal

11 22.9... 12 24...
3 Na 12 Mg
Sodium Alkali Metal Magnesium Alkaline Earth Metal

19 39... 20 40... 21 44... 22 47... 23 50... 24 51... 25 54... 26 55... 27 58... 28 58... 29 63... 30 65.4
4 K Ca Sc Ti V Cr Mn Fe Co Ni Cu Zn
Potassium Alkali Metal Calcium Alkaline Earth Metal Scandium Transition Metal Titanium Transition Metal Vanadium Transition Metal Chromium Transition Metal Manganese Transition Metal Iron Transition Metal Cobalt Transition Metal Nickel Transition Metal Copper Transition Metal Zinc Transition Metal

37 85... 38 87.62 39 88... 40 91.22 41 92... 42 95... 43 96... 44 101.1 45 102... 46 106... 47 107... 48 112... 49 114... 50 118... 51 121... 52 127.6 53 126... 54 131...
5 Rb Sr Y Zr Nb Mo Tc Ru Rh Pd Ag Cd In Sn Sb Te I Xe
Rubidium Alkali Metal Strontium Alkaline Earth Metal Yttrium Transition Metal Zirconium Transition Metal Niobium Transition Metal Molybdenum Transition Metal Technetium Transition Metal Ruthenium Transition Metal Rhodium Transition Metal Palladium Transition Metal Silver Transition Metal Cadmium Transition Metal Indium Post-Transition Metal Tin Post-Transition Metal Antimony Metalloid Tellurium Metalloid Iodine Halogen Xenon Noble Gas

17 35.45
Cl
Chlorine
Halogen

Atomic Number Atomic Mass, u
Symbol
Name
Chemical Group Block
Plot Atomic Mass

5 10.81 6 12.011 7 14.007 8 15.999 9 18.99... 10 20.1...
13 B 14 C 15 N 16 O 17 F 18 Ne
Boron Metalloid Carbon Nonmetal Nitrogen Nonmetal Oxygen Nonmetal Fluorine Halogen Neon Noble Gas

13 26... 14 28... 15 30... 16 32.07 17 35.45 18 39.9
Al Si P S Cl Ar
Aluminum Post-Transition Metal Silicon Metalloid Phosphorus Nonmetal Sulfur Nonmetal Chlorine Halogen Argon Noble Gas

31 69... 32 72... 33 74... 34 78.97 35 79... 36 83...
Ga Ge As Se Br Kr
Gallium Post-Transition Metal Germanium Metalloid Arsenic Metalloid Selenium Nonmetal Bromine Halogen Krypton Noble Gas

of the
Elements

Atomic Number

- Mendeleev's table was based on **atomic masses**. It was the most fundamental property of elements known at the time.
- About 35 years later, the **nuclear atom** was discovered by Ernest Rutherford.
- Henry Moseley developed the concept of atomic number experimentally. The **number of protons** was considered the **basis for the periodic property of elements**.

Periodicity

- Periodicity is the repetitive pattern of a property for elements based on atomic number.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 1.0080 H Hydrogen Nonmetal												5 10.81 B Boron Metalloid	6 12.011 C Carbon Nonmetal	7 14.007 N Nitrogen Nonmetal	8 15.999 O Oxygen Nonmetal	9 18.99... F Fluorine Halogen	10 20.1... Ne Neon Noble Gas
3 7.0 Li Lithium Alkali Metal	4 9.012... Be Beryllium Alkaline Ear...																
11 22.9... Na Sodium Alkali Metal	12 24... Mg Magnesium Alkaline Ear...											13 26.... Al Aluminum Post-Transit...	14 28.... Si Silicon Metalloid	15 30.... P Phosphor... Nonmetal	16 32.07 S Sulfur Nonmetal	17 35.45 Cl Chlorine Halogen	18 39.9 Ar Argon Noble Gas
19 39.... K Potassium Alkali Metal	20 40.... Ca Calcium Alkaline Ear...	21 44.... Sc Scandium Transition M...	22 47.... Ti Titanium Transition M...	23 50.... V Vanadium Transition M...	24 51.... Cr Chromium Transition M...	25 54.... Mn Manganese Transition M...	26 55.... Fe Iron Transition M...	27 58.... Co Cobalt Transition M...	28 58.... Ni Nickel Transition M...	29 63.... Cu Copper Transition M...	30 65.4 Zn Zinc Transition M...	31 69.... Ga Gallium Post-Transit...	32 72.... Ge Germanium Metalloid	33 74.... As Arsenic Metalloid	34 78.97 Se Selenium Nonmetal	35 79.... Br Bromine Halogen	36 83.... Kr Krypton Noble Gas
37 85.... Rb Rubidium Alkali Metal	38 87.62 Sr Strontium Alkaline Ear...	39 88.... Y Yttrium Transition M...	40 91.22 Zr Zirconium Transition M...	41 92.... Nb Niobium Transition M...	42 95.... Mo Molybden... Transition M...	43 96.... Tc Technetium Transition M...	44 101.1 Ru Ruthenium Transition M...	45 102... Rh Rhodium Transition M...	46 106... Pd Palladium Transition M...	47 107... Ag Silver Transition M...	48 112... Cd Cadmium Transition M...	49 114... In Indium Post-Transit...	50 118... Sn Tin Post-Transit...	51 121... Sb Antimony Metalloid	52 127.6 Te Tellurium Metalloid	53 126... I Iodine Halogen	54 131... Xe Xenon Noble Gas

DISPLAY PROPERTY/TREND

Chemical Group Block

Atomic Number

Atomic Mass, u

Symbol

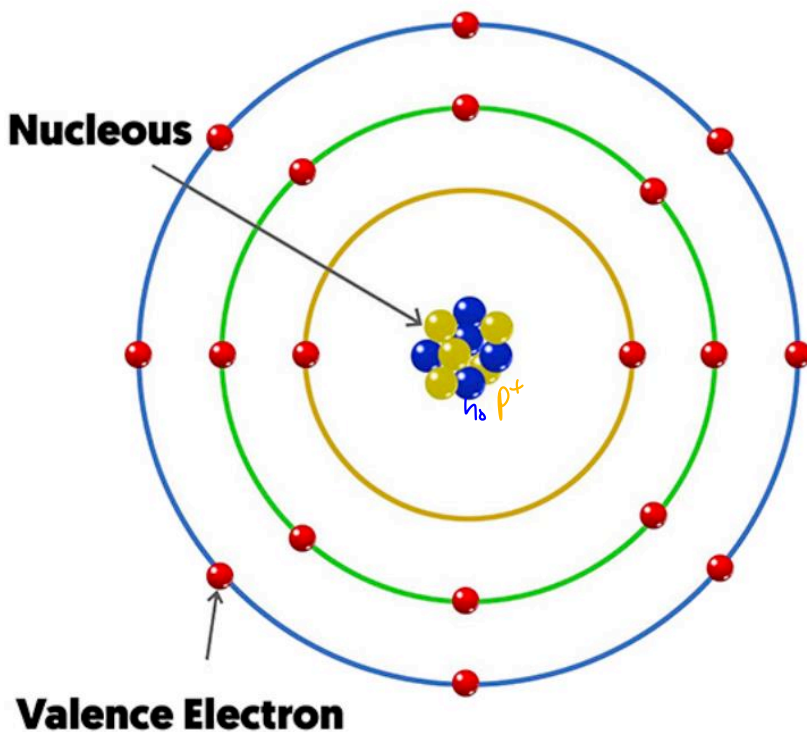
Name

Chemical Group Block

Plot Atomic Mass

of the
Elements

Valence Electrons



Valence electrons are the outermost electrons of an atom that participate in chemical interactions and determine the chemical properties of the element.

They are located in the outermost energy level (valence shell) and play a crucial role in the formation of chemical bonds and chemical reactions.

The **number of valence electrons** is related to the element's position in the periodic table.

The configuration of these electrons influences an atom's ability to gain, lose, or share electrons during chemical interactions.

Valence Electrons

Table 6.3 Electron Configurations of Several Lighter Elements

Element	Total Electrons	Orbital Diagram					Electron Configuration
		1s	2s	2p			3s
Li	3	$\uparrow\downarrow$	$\uparrow$				$1s^2 2s^1$
Be	4	$\uparrow\downarrow$	$\uparrow\downarrow$				$1s^2 2s^2$
B	5	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow$			$1s^2 2s^2 2p^1$
C	6	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow$	$\uparrow$		$1s^2 2s^2 2p^2$
N	7	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow$	$\uparrow$	$\uparrow$	$1s^2 2s^2 2p^3$
Ne	10	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$1s^2 2s^2 2p^6$
Na	11	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$1s^2 2s^2 2p^6 3s^1$

Number of
Valence Electrons

1

2

3

4

5

0

1

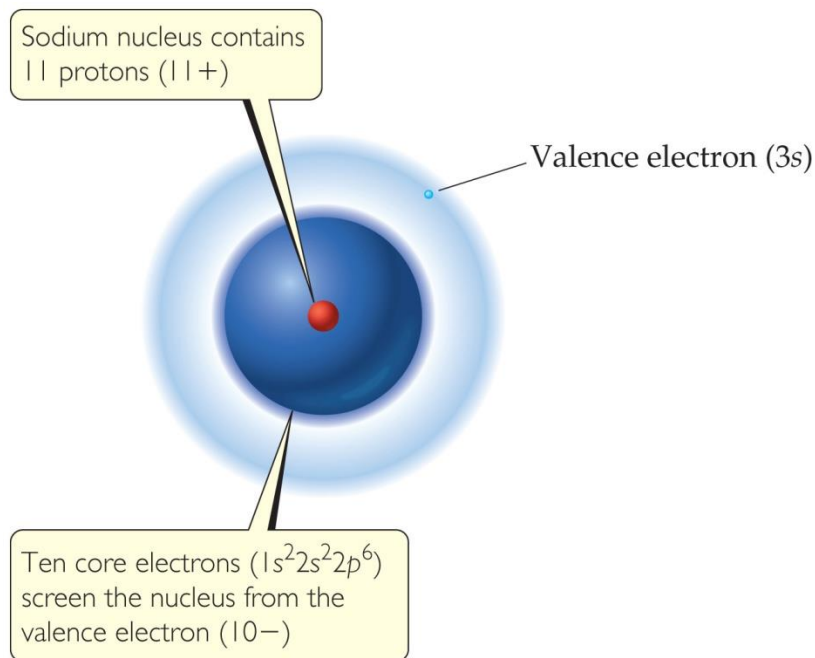
Periodic
Properties
of the
Elements

Effective Nuclear Charge (ENC)

- Many properties depend on attractions between valence electrons and the nucleus.

less valence $e^- \rightarrow e^-$ more attractive

- Electrons are both attracted to the nucleus and repelled by other electrons.
- The forces an electron experiences depend on both factors.



Effective Nuclear Charge

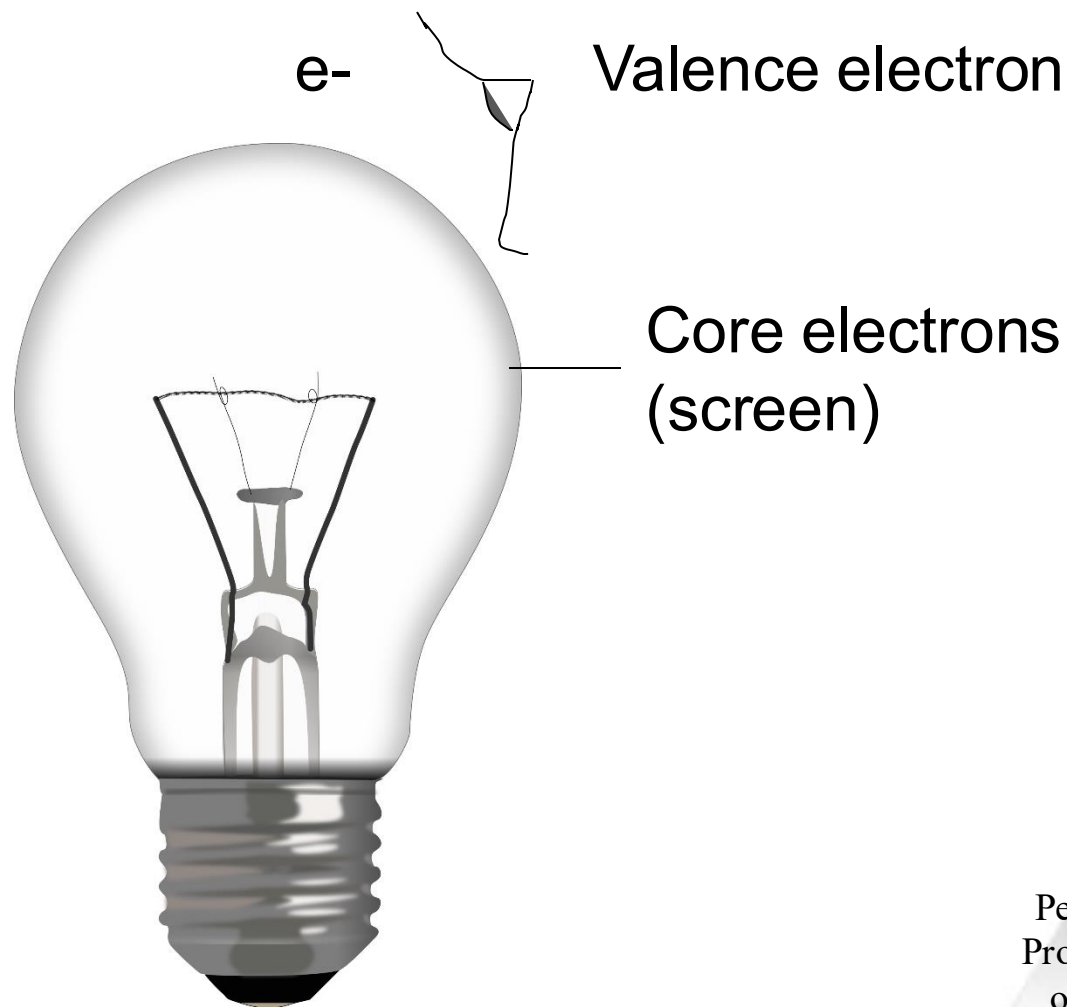
- The effective nuclear charge, Z_{eff} , is found this way:

$$Z_{\text{eff}} = Z - S$$

where Z is the atomic number and S is a screening constant, usually close to the number of inner electrons.

- Effective nuclear charge is a periodic property:
- It increases across a period. S เท่าเดิม Z เพิ่มขึ้น
 - It decreases down a group. ห่างจากนิวเคลียสมากขึ้น $\rightarrow$ ENC ลดลง

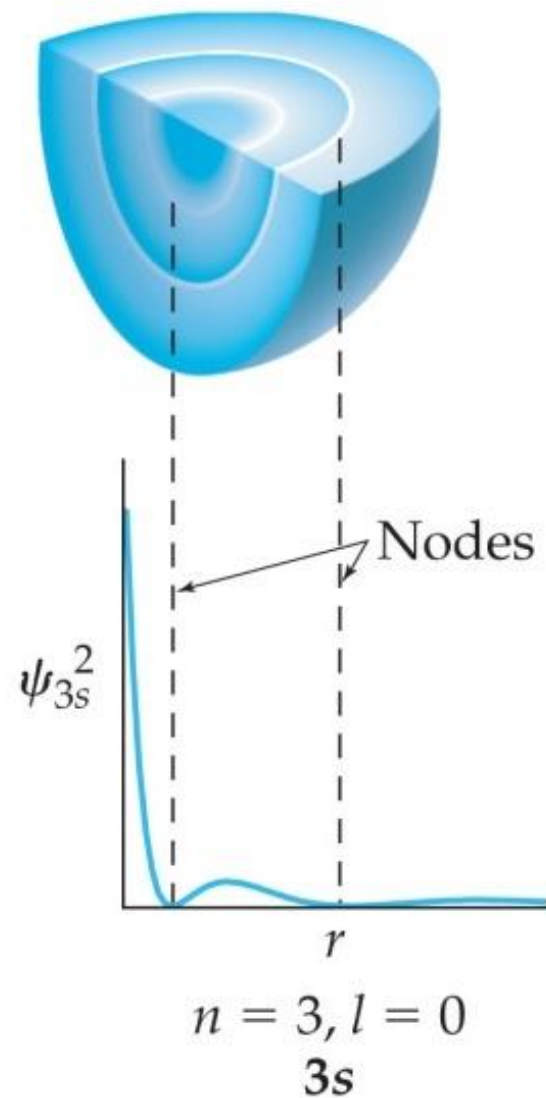
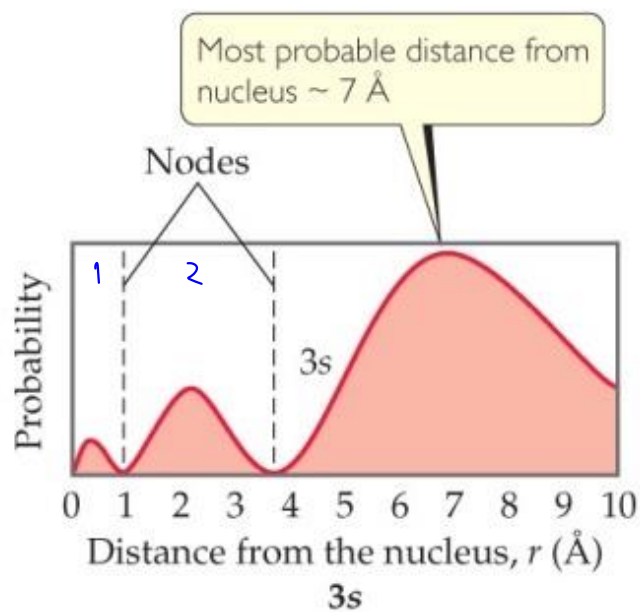
Screening effect



$$Z_{\text{eff}}$$

- Na [Ne]3s¹; $Z = 11 +$
- $Z_{\text{eff}} = 11 - 10 = 1$
- The situation became more complicated if electron in what type of orbital is considered
- For 3s electron in Na, it has smaller probability of being closer to the nucleus
- $Z_{\text{eff}} = 11 - 8.5 = 2.5 +$
may be b/w inner e⁻ / nucleus
Valence e⁻ for any

$$Z_{\text{eff}}$$

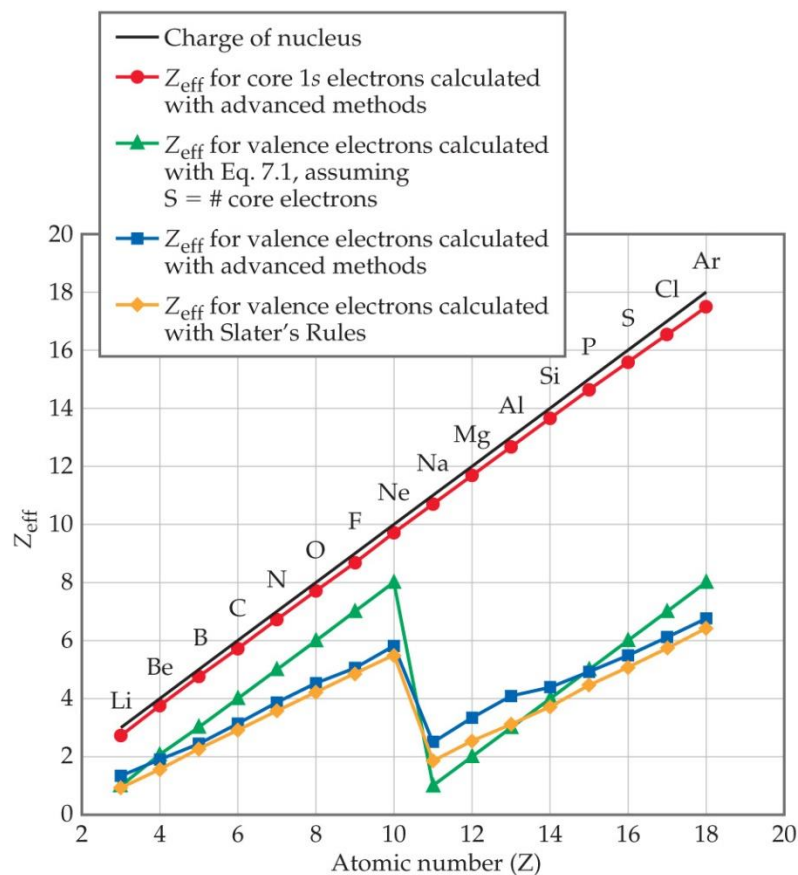


Periodic
Properties
of the
Elements

ENC

- For many-electron atom, the energies of orbitals with same n value increases with l value
- For example in carbon, a $2s$ electron is not screened as much by the core orbitals as is a $2p$ electron.
- $2s$ electron is lower in energy than the $2p$ electron
-

Effective Nuclear Charge Increases across a Period



Effective Nuclear Charge

- The ENC felt by outermost electrons is smaller than that felt by inner electron because of screening effect by the inner electrons
- The big drop (Ne-Na): core electrons are more effective than the valence electrons at screening the nuclear charge

S (Screening constant)

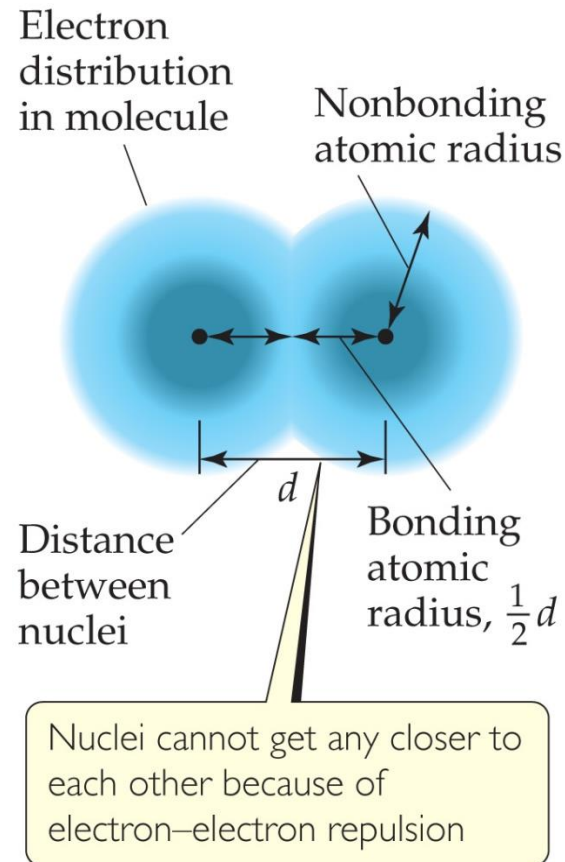
- $Z_{\text{eff}} = Z - S$ (each e- contributes 1.00)
- Simplest assumption is that $S =$ the number of inner electrons

In the case of F: $1s^2 2s^2 2p^5$

- Calculation by $Z_{\text{eff}} = 9 - 2 = 7$

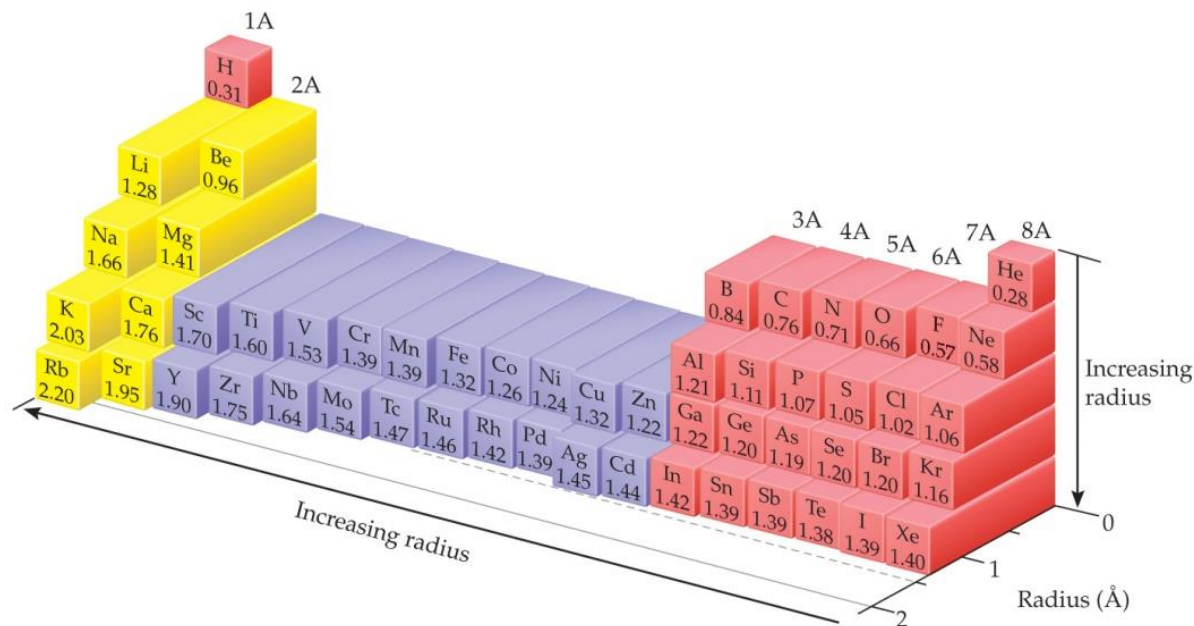
What Is the Size of an Atom?

The **nonbonding atomic radius** or **van der Waals radius** is **half of the shortest distance separating two nuclei during a collision of atoms.**

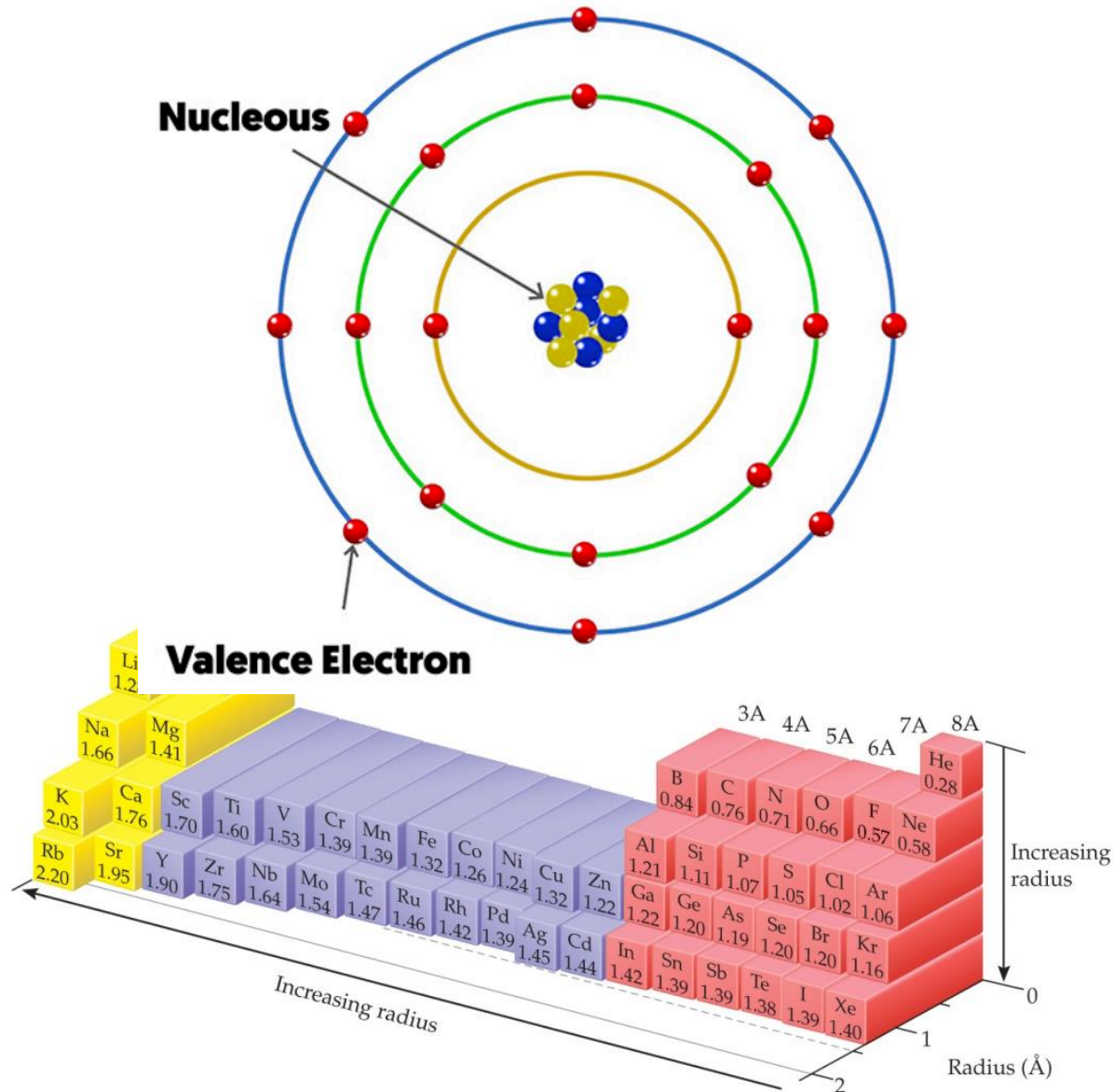


Sizes of Atoms

- The bonding atomic radius is half the internuclear distance when atoms are bonded.
- The bonding atomic radius tends to
 - decrease from left to right across a period ($Z_{\text{eff}} \uparrow$).
 - increase from top to bottom of a group ($n \uparrow$).



Sizes of Atoms



Periodic
Properties
of the
Elements

Sizes of Ions

Group 1A	Group 2A	Group 3A	Group 6A	Group 7A
Li^+ 0.90   Li 1.28	Be^{2+} 0.59   Be 0.96	B^{3+} 0.41   B 0.84	O^{2-} 1.26   O 0.66	F^- 1.19   F 0.57
Na^+ 1.16   Na 1.66	Mg^{2+} 0.86   Mg 1.41	Al^{3+} 0.68   Al 1.21	S^{2-} 1.70   S 1.05	Cl^- 1.67   Cl 1.02
K^+ 1.52   K 2.03	Ca^{2+} 1.14   Ca 1.76	Ga^{3+} 0.76   Ga 1.22	Se^{2-} 1.84   Se 1.20	Br^- 1.82   Br 1.20
Rb^+ 1.66   Rb 2.20	Sr^{2+} 1.32   Sr 1.95	In^{3+} 0.94   In 1.42	Te^{2-} 2.07   Te 1.38	I^- 2.06   I 1.39

 = cation
  = anion
  = neutral atom

- Determined by **interatomic distances in ionic compounds**
- Ionic size depends on
 - the nuclear charge.
 - the number of electrons.
 - the orbitals in which electrons reside.

Sizes of Ions

- Cations are smaller than their parent atoms:
 - The outermost electron is removed and repulsions between electrons are reduced.
- Anions are larger than their parent atoms:
 - Electrons are added and repulsions between electrons are increased.

Group 1A	Group 2A	Group 3A	Group 6A	Group 7A
Li^+ 0.90   Li 1.28	Be^{2+} 0.59   Be 0.96	B^{3+} 0.41   B 0.84	O^{2-} 1.26   O 0.66	F^- 1.19   F 0.57
Na^+ 1.16   Na 1.66	Mg^{2+} 0.86   Mg 1.41	Al^{3+} 0.68   Al 1.21	S^{2-} 1.70   S 1.05	Cl^- 1.67   Cl 1.02
K^+ 1.52   K 2.03	Ca^{2+} 1.14   Ca 1.76	Ga^{3+} 0.76   Ga 1.22	Se^{2-} 1.84   Se 1.20	Br^- 1.82   Br 1.20
Rb^+ 1.66   Rb 2.20	Sr^{2+} 1.32   Sr 1.95	In^{3+} 0.94   In 1.42	Te^{2-} 2.07   Te 1.38	I^- 2.06   I 1.39

 = cation
  = anion
  = neutral atom

Size of Ions— Isoelectronic Series

- In an **isoelectronic series**, ions have the same number of electrons.
- Ionic size decreases with an increasing nuclear charge.
- **An Isoelectronic Series (10 electrons)**
- Note increasing nuclear charge with decreasing ionic radius as atomic number increases

O^{2-}	F^{-}	Na^{+}	Mg^{2+}	Al^{3+}
1.26 Å	1.19 Å	1.16 Å	0.86 Å	0.68 Å

Ionization Energy (I)

- The **ionization energy** is the minimum energy required to remove an electron from the ground state of a gaseous atom or ion.
 - The first ionization energy is that energy required to remove the first electron.
 - The second ionization energy is that energy required to remove the second electron, etc.
- Note: the higher the ionization energy, the more difficult it is to remove an electron!

Ionization Energy

- It requires more energy to remove each successive electron.
- When all valence electrons have been removed, it takes a great deal more energy to remove the next electron.

1st/2nd → 3rd Valence e⁻ removed

Table 7.2 Successive Values of Ionization Energies, I , for the Elements Sodium through Argon (kJ/mol)

Element	I_1	I_2	I_3	I_4	I_5	I_6	I_7
Na	496	4562	(inner-shell electrons)				
Mg	738	1451	7733				
Al	578	1817	2745	11,577			
Si	786	1577	3232	4356	16,091		
P	1012	1907	2914	4964	6274	21,267	
S	1000	2252	3357	4556	7004	8496	27,107
Cl	1251	2298	3822	5159	6542	9362	11,018
Ar	1521	2666	3931	5771	7238	8781	11,995

Periodic Trends in First Ionization Energy (I_1)

- 1) I_1 generally increases across a period.
- 2) I_1 generally decreases down a group.
- 3) The *s*- and *p*-block elements show a larger range of values for I_1 . (The *d*-block generally increases slowly across the period; the *f*-block elements show only small variations.)

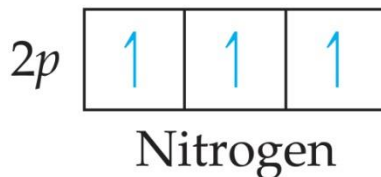
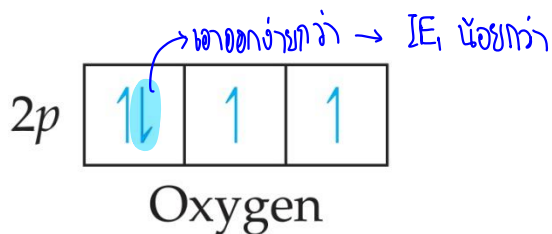
IE

- IE



Irregularities in the General Trend

- The trend is **not** followed when the added valence electron in the next element
 - enters a new sublevel (higher energy sublevel);
 - is the **first** electron to pair in one orbital of the sublevel (electron repulsions lower energy).

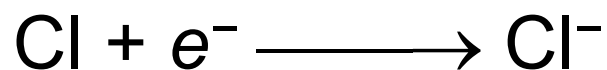


Electron Configurations of Ions

- **Cations:** The electrons are lost from the *highest energy level* (n value).
 - Li^+ is $1s^2$ (losing a $2s$ electron).
 - Fe^{2+} is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$ (losing two $4s$ electrons).
- **Anions:** The electron configurations are filled to $ns^2 np^6$; e.g., F^- is $1s^2 2s^2 2p^6$ (gaining one electron in $2p$).

Electron Affinity

- **Electron affinity** is the energy change accompanying the **addition of an electron to a gaseous atom:**



gaseous state

IE = พลังงานที่ต้องใช้เพื่อเอา e⁻ ออก
EA = พลังงานที่ปล่อยออกมาเมื่อ e⁻ เข้าไป

- It is typically **exothermic**, so, for most elements, it is **negative!**

release heat to environment

General Trend in Electron Affinity

- Not much change in a group.
- Across a period, it generally increases. *Three* notable exceptions include the following:

- 1) Group 2A: s sublevel is full!
- 2) Group 5A: p sublevel is half-full!
- 3) Group 8A: p sublevel is full!

Note: the electron affinity for many of these elements is *positive* (X^- is unstable).

1A							8A
H -73							He > 0
	2A	3A	4A	5A	6A	7A	
Li -60	Be > 0	B -27	C -122	N > 0	O -141	F -328	Ne > 0
Na -53	Mg > 0	Al -43	Si -134	P -72	S -200	Cl -349	Ar > 0
K -48	Ca -2	Ga -30	Ge -119	As -78	Se -195	Br -325	Kr > 0
Rb -47	Sr -5	In -30	Sn -107	Sb -103	Te -190	I -295	Xe > 0

2s orbital filled